Amirkabir University of Technology (Tehran Polytechnic)

Department of Mechanical Engineering · Turbomachinery Course Project

Redesign of the Mitsubishi Power M701JAC Axial Compressor

Report 2 — Thermodynamic and Mean-Line Aerodynamic Design

448 MW ISO rating · 15 stages · pressure ratio 25 · 3000 rpm

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Prepared with the assistance of Claude (Anthropic); see Section 11. All calculations are reproducible with `m701jac/report2/report2_calc.py` *in the project repository.*

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1 Selection of the M701JAC

The Phase 1 proposal took a 1990s-generation heavy-duty compressor (MAPNA MGT-70, pressure ratio 12 in 16 stages) as its subject. For the redesign we moved to the **Mitsubishi Power M701JAC**, because it represents the current limit of what an industrial axial compressor is asked to do, for the following reasons:

- **Highest efficiency class.** 44.0 % simple-cycle and above 64 % combined-cycle efficiency (LHV) [D2] place it among the most efficient heavy-duty gas turbines. Since the compressor absorbs roughly half of the turbine work (Section 5), such efficiency is only possible with a very good compressor.
- **Very high stage loading.** A pressure ratio of 25 in only 15 stages means an average stage pressure ratio of 1.24, against 1.17 in the Phase 1 machine. This requires transonic front stages, three-dimensional blading and boundary-layer control [D6–D8]. These are exactly the gas-dynamic problems this course studies.
- **The compressor enables the hot section.** The 1650 °C-class turbine inlet temperature raises the exhaust temperature unless the pressure ratio rises with it. The compressor was redesigned from PR 23 to PR 25 precisely to hold the exhaust temperature down [D4]. The compressor is therefore the key to the cycle, not a passive component.
- **Operability.** Variable inlet guide vanes plus three rows of variable stator vanes control rotating stall at start-up and keep the exhaust temperature high at part load. The machine has 50 % turndown, a 53 MW/min ramp rate and a 30-minute start [D1, D6]. These are the requirements of a modern grid with a large renewable share.
- **Suited to the Iranian grid.** It is a 50 Hz machine with direct drive at 3000 rpm and no gearbox. It burns natural gas or oil [D3], and Mitsubishi offers a conversion path to hydrogen firing [D2].
- **Proven reliability.** The J-series fleet has more than three million operating hours [D2]; ten M701JAC units in Thailand alone total 6,700 MW [D3].

2 The machine and how its compressor works

The working principle and theory follow Report 1. The main data of the M701JAC are [D2, D5, D6]:

Table 1. Main data of the M701JAC.

Item	M701JAC (448 MW rating)
Output / efficiency (ISO, LHV)	448 MW / 44.0 % (heat rate 8182 kJ/kWh)
Compressor	Axial, 15 stages, 3-D blading; variable IGV + variable stator vanes on stages 1–3
Total pressure ratio	25
Speed	3000 rpm (50 Hz, cold-end direct drive)
Exhaust flow / temperature	765 kg/s / 663 °C
Turbine	4 stages, all blade and vane rows air-cooled; 1650 °C class
Published compressor features	transonic front stages (shock-loss reduction); start-up bleeds at low, middle and high pressure; rotating stall disappears at ≈ 60 % speed

Each stage is a rotor that adds swirl, and therefore work, followed by a stator that removes it. Both rows act as diffusers: the rotor in the relative frame ($W_2 < W_1$) and the stator in the absolute frame ($V_3 < V_2$). The relations used below are those of Report 1:

$$\frac{T_{02}}{T_{01}} = r_p^{\frac{\gamma-1}{\gamma\eta_{pc}}}, \eta_{isc} = \frac{r_p^{(\gamma-1)/\gamma} - 1}{r_p^{(\gamma-1)/(\gamma\eta_{pc})} - 1} \quad (1)$$

$$\Delta h_0 = U\,\Delta V_w, \frac{V_a}{U} = \phi, \frac{\Delta h_0}{U^2} = \psi \quad (2)$$

$$R = \frac{h_2 - h_1}{h_3 - h_1} = 0.5 \Rightarrow V_{w1} = \frac{U - \Delta V_w}{2}, V_{w2} = V_{w1} + \Delta V_w, W_1 = V_2, W_2 = V_1 \quad (3)$$

$$T = T_0 - \frac{V^2}{2c_p}, p = p_0\left(\frac{T}{T_0}\right)^{\frac{\gamma}{\gamma-1}}, M_{rel} = \frac{W}{\sqrt{\gamma R T}} \quad (4)$$

$$\text{de Haller: } \frac{W_2}{W_1}(\text{rotor}), \frac{V_3}{V_2}(\text{stator}) : \geq 0.75 \text{ light}, 0.70 - 0.75 \text{ heavy}, < 0.70 \text{ not acceptable} \quad (5)$$

3 Assumed values

Every value below is either an ISO or physical standard, a common-practice value, or a design choice made in this report. **Nothing else enters the calculations.**

Table 2. All assumed values used in this report.

Quantity	Value	Type	Use / remark
Inlet total state T_{01}, p_{01}	288.15 K, 101.325 kPa	ISO standard	Basis of the catalogue rating
Air: γ, c_p, R	1.4, 1004.5 J/kg·K, 287 J/kg·K	Standard value	Perfect gas throughout
Combustion gas: γ_g, c_{pg}	1.333, 1148 J/kg·K	Common practice	Turbine and exhaust in the cycle check only

Quantity	Value	Type	Use / remark
Fuel lower heating value	50.0 MJ/kg	Standard (methane)	Fuel and air flows (±4 % LHV changes air flow by ±0.1 %)
Generator + mechanical efficiency	0.985	Common practice	Shaft power for the cycle check
Combustor pressure loss	4 % of p_{02}	Common practice (3–5 %)	Turbine pressure ratio
Turbine exit pressure	1.02 × ambient	Common practice	Exhaust diffuser and duct
Turbine polytropic efficiency η_{pe}	0.90–0.92	Common practice	Cooled turbine at its practical best; *fixes* η_{pc} (Section 5)
Compressor polytropic efficiency η_{pc}	**0.86**	Derived (Section 5)	Inside the band 0.84–0.875 implied by the catalogue
Axial velocity V_a	170 m/s, constant	Design choice	Inlet axial Mach 0.51; constant through all stages
First-rotor hub-to-tip ratio	0.50	Design choice	Typical heavy-duty first stage (0.40–0.55)
Degree of reaction	0.5 at mean radius, all stages	Design choice	Symmetric velocity triangles
Stage work distribution	equal, or non-uniform (stage 1 = 0.85, stage 15 = 0.80 × mean)	Design choice	Both are calculated (Section 8)
Annulus shape	constant mean or constant tip diameter	Design choice	Both are calculated (Section 8)
Radial whirl distribution	free vortex ($V_w r$ = const)	Common practice	Tip relative Mach number only
de Haller limits	0.75 / 0.70	Common practice	Equation (5)
Blade density, taper factor	7800 kg/m³, 0.6	Common practice	Stage-1 root stress only

4 Design basis and air flow

The catalogue fixes the heat input from the heat rate, and the fuel and air flows follow:

$$\dot{Q} = P \cdot HR = 448 \times \frac{8182}{3600} = 1018.2\,\text{MW}, \dot{m}_f = \frac{\dot{Q}}{LHV} = 20.36\,\text{kg/s}, \dot{m}_a = 765 - 20.36 = 744.6\,\text{kg/s} \quad (6)$$

The compressor must therefore deliver **744.6 kg/s at a total-pressure ratio of 25**. The overall energy balance $\dot{Q} - \dot{m}_{exh} c_{pg} \left(T_{exh} - T_{01} \right)$ = 449.1 MW of shaft power, against 448/0.985 = 454.8 MW required. It closes to within 0.6 % of the heat input, so the catalogue data are consistent.

5 Cycle and compressor efficiency

The cycle is evaluated as follows:

$$T_{02} = T_{01} r_p^{\frac{\gamma - 1}{\gamma \eta_{pc}}}, \dot{W}_C = \dot{m}_a c_p \left(T_{02} - T_{01} \right), \dot{m}_g c_{pg} \left(T_{03} - T_{01} \right) = \dot{m}_a c_p \left(T_{02} - T_{01} \right) + \dot{Q} \quad (7)$$

$$\frac{T_{03}}{T_{04}} = \left(\frac{p_{03}}{p_{04}} \right)^{\frac{\gamma_g - 1}{\gamma_g} \eta_{pe}}, T_{04} = 936.15\,\mathrm{K}, \frac{p_{03}}{p_{04}} = \frac{25 \times 0.96}{1.02} = 23.53 \quad (8)$$

T_{03} is the equivalent mixed turbine-inlet temperature, with the cooling air included. For any assumed compressor efficiency, equation (7) gives T_{02} and T_{03}, and equation (8) then gives the turbine efficiency that the catalogue exhaust temperature requires:

Table 3. Cycle closure for a range of compressor efficiencies (448 MW rating).

η_pc	T_{02} [K]	$\dot{W}$_C [MW]	T_{03} [°C]	Implied η_pe	η_isc	$\dot{W}$_C / $\dot{W}$_T
0.82	884.5	446.1	1682	0.934	0.729	0.498
0.84	861.2	428.6	1662	0.921	0.759	0.488
0.86	**839.6**	**412.4**	**1644**	**0.909**	**0.788**	**0.479**
0.88	819.4	397.4	1627	0.897	0.818	0.469
0.90	800.6	383.3	1611	0.886	0.848	0.460

A turbine has few stages, accelerating flow and no stall, so its efficiency is close to the best attainable. For a cooled heavy-duty turbine that is η_{pe} = 0.90–0.92. Table 3 then fixes the compressor at η_{pc} = **0.84–0.875**, clearly lower than the turbine, as expected for a decelerating, stall-limited machine. We adopt η_{pc} **= 0.86**. Two independent checks support it:

- the implied turbine efficiency is 0.909;
- the mixed turbine-inlet temperature is 1644 °C, matching the published 1650 °C class [D3, D4].

Compressor result: delivery temperature T_{02} = **839.6 K (566 °C)**, isentropic efficiency **0.788**, shaft power **412 MW**. The compressor absorbs **48 %** of the turbine work.

Inlet and outlet states. Figure 1 shows the whole compression on the T–s diagram with its four end states:

- inlet stagnation state 01: 288.15 K, 101.3 kPa;
- inlet static state 1, at the IGV inlet: 273.8 K, 84.7 kPa;
- outlet stagnation state 02: 839.6 K, 2533 kPa;
- outlet static state 2, at the compressor exit: 825.2 K, 2384 kPa.

Each static state lies $V_a^2 / 2 c_p$ = 14.4 K below its stagnation state, at the same entropy. The entropy rise $s_{02} - s_{01} = c_p \ln \left(T_{02} / T_{01} \right) - R \ln \left(p_{02} / p_{01} \right)$ = 150.4 J/kg·K measures the losses. The isentropic end state 02s (722.8 K) would need 436.6 kJ/kg instead of 553.9 kJ/kg.

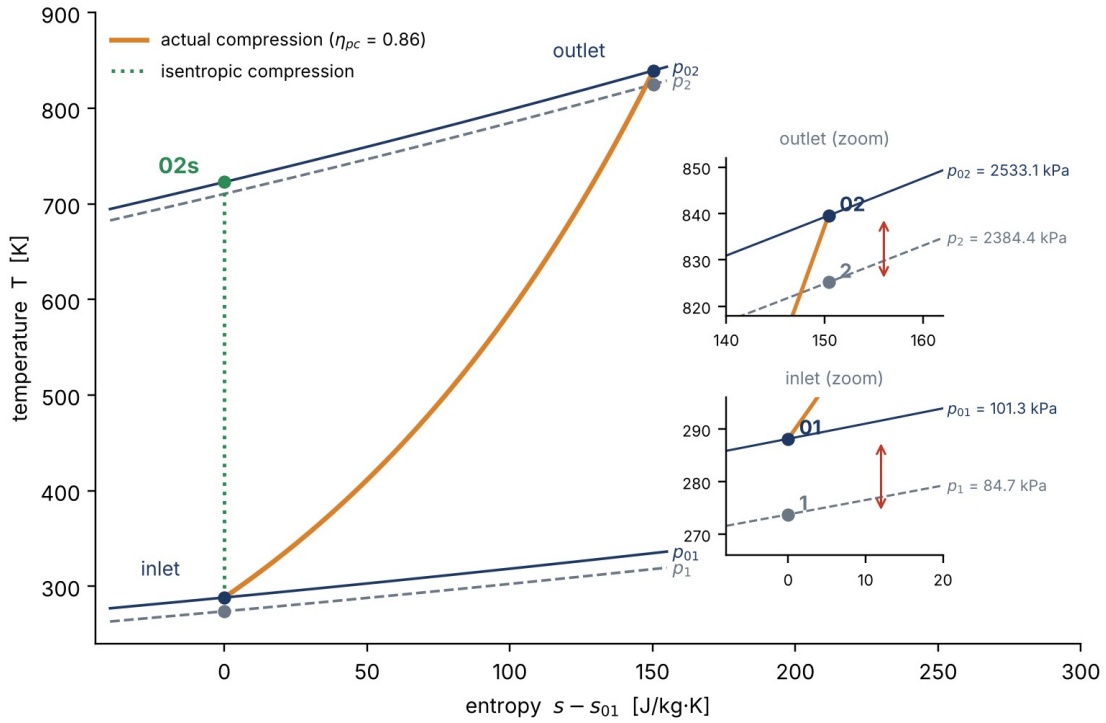


Figure 1. T–s diagram of the whole compressor: inlet and outlet stagnation states (01, 02) and static states (1, 2), with the isentropic end state 02s. Each static state lies 14.4 K below its stagnation state (zoomed panels).

Real-gas check of the delivery temperature. With constant c_p (γ = 1.4) the delivery temperature is 839.6 K. Repeating the compression with temperature-dependent properties of humid ISO air (Cantera, GRI-Mech 3.0 data) gives 811.4 K, about 28 K lower. Table 4 separates the effects.

Table 4. Delivery temperature with constant and with temperature-dependent properties (η_{pc} = 0.86, r_p = 25).

	Constant c_p (γ = 1.4)	Variable c_p, dry air	Variable c_p, humid ISO air
c_p at inlet / outlet [J/kg·K]	1004.5 / 1004.5	1002.3 / 1100.9	1007.7 / 1107.2
γ at inlet / outlet	1.400 / 1.400	1.401 / 1.353	1.401 / 1.352
Isentropic end temperature T_{02s} [K]	722.8	707.8	706.7
Delivery temperature T_{02} [K]	**839.6**	812.8	**811.4**
Work Δh_0 [kJ/kg]	553.9	547.9	549.4
Isentropic efficiency	0.788	0.792	0.792
Effective constant γ giving the same T_{02}	1.400	1.383	1.382

The difference comes almost entirely from c_p; humidity accounts for only 1.4 K. For variable c_p the polytropic relation reads

$$\int_{T_{01}}^{T_{02}} \frac{c_p(T)}{T}\,dT = \frac{R}{\eta_{pc}} \ln r_p \quad (9)$$

The right-hand side depends only on the pressure ratio and the efficiency. Between inlet and outlet c_p rises by 10 % (1008 → 1107 J/kg·K) and γ falls from 1.40 to 1.35. A larger c_p makes the left-hand side grow faster with temperature, so the same value is reached at a lower T_{02}. In the constant-γ formula this appears as a smaller exponent $(\gamma-1)/(\gamma\,\eta_{pc})$: an effective γ of 1.382 reproduces the real-gas result. The work, by contrast, hardly changes. Along a polytropic path $dh = v\,dp/\eta_{pc}$, so

$$\Delta h_0 = \frac{R}{\eta_{pc}} \int T\, d(\ln p) \quad (10)$$

This depends on the temperature along the path, which is only slightly lower. The 5 % smaller temperature rise is offset by a 4.5 % higher mean c_p, so the work falls by only 0.8 % (553.9 → 549.4 kJ/kg).

Consequences. Work, power, velocity triangles and de Haller numbers are essentially unchanged. Quantities that depend directly on temperature are affected:

- the delivery air is 28 K cooler, which matters for the turbine cooling air and for the combustor temperature rise;
- the exit density is about 3.5 % higher, so the last-stage blade height becomes about 59 mm instead of 61 mm.

Perfect-gas values are kept in the rest of the report; the real-gas delivery temperature is also listed in Table 8.

6 Mean-line sizing

Inlet annulus. At the IGV inlet the flow is axial, with V_a = 170 m/s:

$$T_1 = 288.15 - \frac{170^2}{2\times 1004.5} = 273.76\,\mathrm{K},\; p_1 = 84.70\,\mathrm{kPa},\; \rho_1 = 1.078\,\mathrm{kg/m^3},\; A_1 = \frac{\dot m_a}{\rho_1 V_a} = 4.063\,\mathrm{m^2}$$

With a hub-to-tip ratio of 0.50: D_t = 2.626 m, D_h = 1.313 m, D_m = 1.970 m, and the inlet axial Mach number is 0.513.

Loading. The total work is $\Delta h_0 = c_p(T_{02} - T_{01})$ = 553.9 kJ/kg, which is **36.93 kJ/kg per stage** on average. With a constant mean diameter of 1.970 m, $U_m = \pi D_m N/60$ = 309.4 m/s, ϕ = 0.549 and

$$\psi_{total} = \frac{\Delta h_0}{U_m^2} = \frac{553.9\times 10^3}{309.4^2} = 5.786,\; \psi_{stage} = \frac{5.786}{15} = 0.386,\; \Delta V_w = \frac{36.93\times 10^3}{309.4} = 119.3\,\mathrm{m/s}$$

A stage-loading coefficient of 0.386 is high for an industrial compressor. It confirms the “high-load” character of this machine [D3].

7 Stage 1: velocity triangles and gas dynamics

The design case selected in Section 8 has a constant tip diameter, which puts the stage-1 mean diameter at 1.923 m (U = 302.1 m/s). With about 0.85 × mean work in stage 1 (Δh_0 = 31.6 kJ/kg) and R = 0.5, equations (2)–(4) give:

Table 5. Stage 1 at mean radius (design case D).

Quantity	Value	Quantity	Value
U, V_a, ϕ	302.1 m/s, 170 m/s, 0.563	$\psi, \Delta V_w$	0.346, 104.5 m/s
V_{w1} / V_{w2}	98.8 / 203.3 m/s	$\alpha_1 = \beta_2 / \alpha_2 = \beta_1$	30.2° / 50.1°
$V_1 = W_2$	196.6 m/s	$V_2 = W_1$	265.0 m/s
Static T_1, p_1	268.9 K, 79.55 kPa	M_1 (absolute)	0.598
M_{rel} at mean radius	0.806	M_{rel} at tip (free vortex)	**1.157**
de Haller, rotor = stator	**0.742**	Stage pressure ratio	1.365

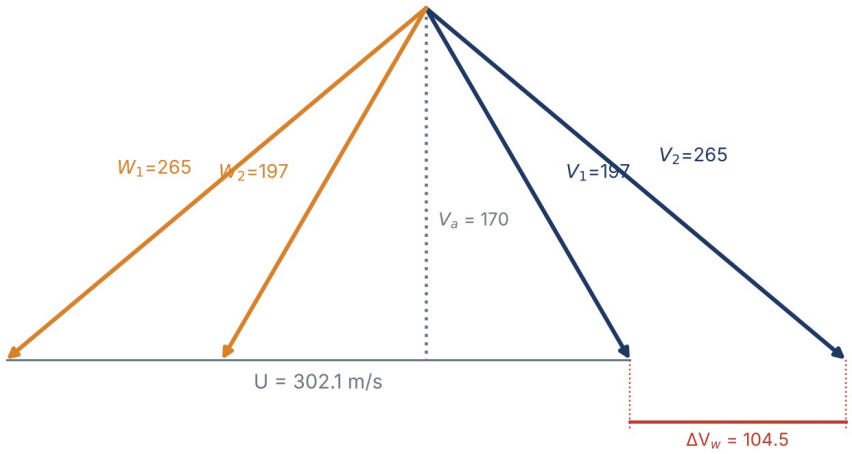


Figure 2. Stage-1 velocity triangles at mean radius (design case D).

Two results stand out:

- **The tip is transonic** (M_{rel} = 1.16 at the tip, 0.81 at mean radius). This is why MHI uses three-dimensional design to reduce shock losses in the initial stages [D6]. It is also why the first stage cannot carry full load.
- **The de Haller number of 0.742 lies in the heavy-loading band** (0.70–0.75). Such a stage needs advanced blade design. This agrees with the published description of a high-loading, boundary-layer-controlled compressor [D7, D8].

8 Stage-by-stage design

The split of work between stages and the shape of the annulus are design decisions. Two work distributions (equal and non-uniform) and two annulus shapes (constant mean and constant tip diameter) give four design variants, each reproducing PR 25 and 744.6 kg/s exactly:

Table 6. Design variants: work distribution and annulus shape.

Case	Work distribution	Annulus	Max ψ	Min de Haller (stage)	Stage PR 1 / 15	Blade height 1 / 15	Stage-1 tip M_rel
A	equal	constant D_m = 1.970 m	0.386	0.712 (all)	1.435 / 1.144	684 / 78 mm	1.173
B	non-uniform	constant D_m	0.411	**0.697 (8)**	1.365 / 1.114	690 / 77 mm	1.160
C	equal	constant D_t = 2.626 m	0.401	0.708 (1)	1.435 / 1.144	696 / 62 mm	1.171
D	**non-uniform**	**constant D_t**	**0.346**	**0.732 (2)**	**1.365 / 1.114**	**703 / 61 mm**	**1.157**

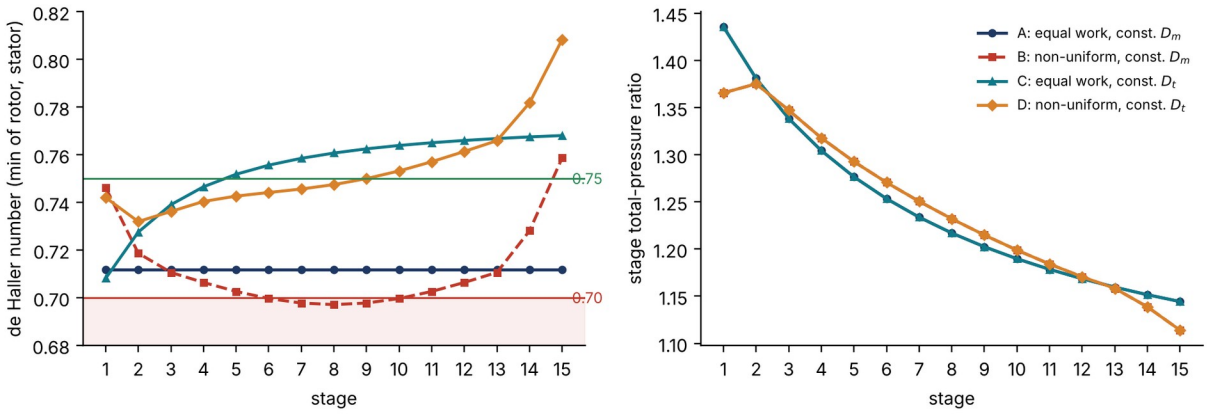


Figure 3. de Haller number and stage pressure ratio for the four cases.

What the comparison shows:

- **Equal work is not equal pressure ratio.** Even with equal work the stage pressure ratio falls from 1.435 to 1.144, because each stage starts hotter (cases A and C).
- **Non-uniform work at constant mean diameter fails** (case B). Moving work to the middle stages at constant blade speed pushes stages 6–10 below de Haller 0.70.
- **A constant tip diameter makes the non-uniform distribution work** (case D). The hub rises, so the mean radius and the blade speed grow from 302 to 403 m/s along the machine. The middle and rear stages can then take more work at a *lower* loading coefficient. The first two stages stay limited by the transonic tip and remain the most heavily loaded aerodynamically.
- **Case D is adopted as the redesign:** minimum de Haller 0.732, no stage below 0.70, and a loading that falls smoothly from 0.35 to 0.18.

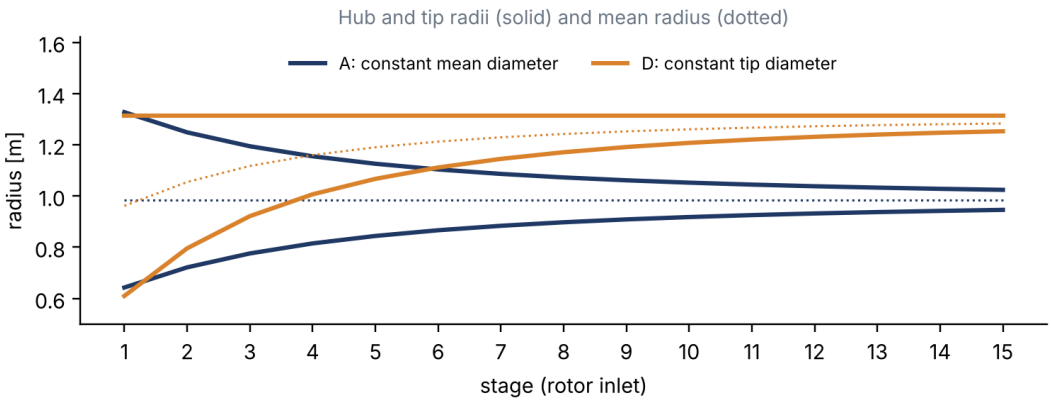


Figure 4. Hub, mean and tip radii along the compressor for constant mean diameter (A) and constant tip diameter (D).

Table 7. The redesigned compressor, stage by stage (case D, rotor-inlet values at mean radius; f = stage work / mean stage work; M tip = relative Mach number at the blade tip; dH = de Haller number).

Stage	f	ΔT_0 [K]	Stage PR	p_0 out [bar]	T_0 out [K]	D_m [m]	h [mm]	U [m/s]	φ	ψ	α_1/α_2 [°]	M tip	dH
1	0.855	31.4	1.365	1.38	319.6	1.923	703	302.1	0.563	0.346	30.2/50.1	1.157	0.742
2	0.970	35.7	1.375	1.90	355.2	2.109	518	331.2	0.513	0.327	33.3/52.3	1.053	0.732
3	1.005	37.0	1.347	2.56	392.2	2.234	392	351.0	0.484	0.301	35.8/53.3	0.962	0.736
4	1.024	37.7	1.318	3.38	429.8	2.320	307	364.4	0.467	0.285	37.5/54.0	0.889	0.740
5	1.041	38.3	1.293	4.37	468.1	2.380	246	373.9	0.455	0.275	38.6/54.5	0.831	0.743
6	1.054	38.8	1.271	5.55	506.9	2.424	202	380.8	0.446	0.268	39.3/54.9	0.784	0.744
7	1.063	39.1	1.250	6.93	545.9	2.458	168	386.1	0.440	0.263	39.9/55.1	0.743	0.746
8	1.066	39.2	1.232	8.54	585.1	2.484	142	390.2	0.436	0.258	40.4/55.3	0.708	0.747
9	1.063	39.1	1.215	10.38	624.2	2.504	122	393.4	0.432	0.254	40.8/55.4	0.677	0.750
10	1.054	38.8	1.199	12.44	662.9	2.520	106	395.9	0.429	0.248	41.2/55.5	0.650	0.753
11	1.041	38.3	1.184	14.73	701.2	2.534	93	398.0	0.427	0.243	41.6/55.5	0.626	0.757
12	1.024	37.7	1.170	17.24	738.8	2.544	82	399.6	0.425	0.237	41.9/55.5	0.604	0.761
13	1.005	37.0	1.158	19.97	775.8	2.553	74	401.0	0.424	0.231	42.2/55.4	0.585	0.766
14	0.930	34.2	1.139	22.74	810.0	2.560	66	402.1	0.423	0.212	43.0/55.1	0.564	0.782
15	0.804	29.6	1.114	25.33	839.6	2.566	61	403.0	0.422	0.183	44.1/54.5	0.542	0.808

9 Mechanical note

The first rotor carries the largest centrifugal load because it has the largest annulus area. With $\sigma = K\,\rho_b\,\omega^2\,A_1/2\pi$ the root stress is about **300 MPa** for a tapered steel blade, an acceptable level for compressor blade steels. The 50 Hz rotational frequency and its multiples, together with the stator blade-passing frequencies, are the excitations against which the front blades must be detuned [D6]. A detailed vibration analysis is outside the scope of this report.

10 Results and conclusions

Table 8. Summary of the redesign.

Quantity	Redesign (case D)	Catalogue / published
Air mass flow	744.6 kg/s	765 kg/s exhaust − fuel
Total pressure ratio	25.0 (15 stages)	25
Delivery temperature	839.6 K (566 °C); 811 K with variable c_p	—
Compressor power / share of turbine work	412 MW / 48 %	—
Polytropic / isentropic efficiency	0.86 / 0.788	—
Equivalent turbine inlet temperature	1644 °C	1650 °C class
Stage-1 tip diameter / tip relative Mach	2.63 m / 1.16	transonic front stages
Stage pressure ratio, first → last	1.365 → 1.114	—
Blade height, first → last	703 → 61 mm	reduced outlet area
Lowest de Haller number	0.732 (stage 2)	high-load design

Conclusions:

1. The catalogue data close to within 0.6 %, so the redesign targets (744.6 kg/s, PR 25) are firm.
2. The turbine runs near its best attainable efficiency, so the catalogue fixes the compressor polytropic efficiency at 0.84–0.875. With 0.86 the machine delivers 840 K air and absorbs 412 MW, about half of the turbine work.
3. Equal pressure ratio per stage is not physical. Equal work already gives 1.43 → 1.14, and a realistic non-uniform distribution gives 1.37 → 1.11.
4. Of the four design variants, only the constant-tip annulus allows the non-uniform work distribution within the de Haller limits. It is adopted as the redesign.
5. The first two stages are transonic at the tip and heavily loaded (de Haller 0.73–0.74), which explains the three-dimensional, high-loading design of the real machine.

Next steps (Report 3): blade numbers and chords from diffusion-factor limits; radial variation in stages 1–3; the role of the variable IGV and stators at off-design; the standard test and the virtual test.

11 Use of AI

All calculations in this report were coded and run by Claude (Anthropic) under the team's direction. Every assumed value is listed in Table 2 and was reviewed by the team. The AI also checked the catalogue data for consistency and compared the perfect-gas results with a real-gas calculation (Section 5).

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